Amir

AI Researcher & Systems Engineer

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Core Expertise

- **Active Research:** Large Language Models (LLMs), Reinforcement Learning from Human Feedback (RLHF), Distributed Training, Transformer Architectures.
- **Core Production:** Python, C++, Rust, Go, CUDA, PyTorch, JAX, TensorRT, Triton.
- **Tooling & Systems:** Kubernetes, Docker, Ray, CI/CD, Apache Kafka, gRPC, PostgreSQL.

Experience & Research

Senior AI Systems Engineer

Tech Nova Labs | San Francisco, CA

June 2021 - Present

- Engineered a distributed training pipeline using PyTorch and Ray, accelerating LLM pre-training by 35% across 512 A100 GPUs.
- Architected a low-latency inference engine leveraging TensorRT and CUDA custom kernels -> Achieved sub-50ms latency for a 7B parameter model under high concurrency.
- Designed and deployed robust API gateways with Go and gRPC -> Scaled to support over 100K requests per second with 99.99% uptime.

Machine Learning Researcher

Quantum Cognitive Institute | Boston, MA

August 2018 - May 2021

- Investigated novel self-attention mechanisms -> Published findings at NeurIPS 2020 demonstrating a 15% reduction in memory overhead.
- Developed an automated hyperparameter optimization framework utilizing Bayesian optimization -> Reduced experiment iteration times by 40%.
- Curated and pre-processed a multi-modal dataset comprising 10TB of text and image pairs -> Enabled the training of foundational multimodal models.

Selected Projects & System Architectures

Project Valhalla: Decentralized Federated Learning

- Open-source framework written in Rust and Python for privacy-preserving distributed learning.
- Implemented secure multi-party computation (SMPC) protocols to encrypt model gradients -> Ensured zero knowledge leakage during client-server aggregation.
- **Tech Stack:** Rust, PyTorch, gRPC, Cryptography.

NeuralSynth: Real-Time Audio Generation

- Developed an end-to-end generative model for real-time procedural audio synthesis using C++ and JAX.
- Optimized the synthesis graph for embedded devices -> Reached real-time synthesis capabilities on ARM Cortex-M microcontrollers.
- **Tech Stack:** C++, JAX, WebAssembly, DSP.

Education

Master of Science in Computer Science

Massachusetts Institute of Technology (MIT) | *2016 - 2018*

- Specialization: Artificial Intelligence and Distributed Systems.
- Key Courses: Advanced Machine Learning, Parallel Computing, Probabilistic Graphical Models.
- Thesis: “Optimizing Communication Bottlenecks in Decentralized Deep Learning Networks”

Bachelor of Science in Mathematics and Computer Science

Stanford University | *2012 - 2016*

- Graduated with Honors.
- Key Courses: Real Analysis, Linear Algebra, Data Structures and Algorithms, Operating Systems.